

MODULAR MULTI-AGENT DOCUMENT AI

Project ID: 25-26J-129

Project Proposal Report

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DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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ABSTRACT

This research focuses on the integration of **Explainable Artificial Intelligence (XAI)** and **Human-in-the-Loop (HITL)** mechanisms within a multi-agent document AI platform. The objective is to enhance trust, accountability, and reliability by ensuring that system outputs are transparent and subject to expert oversight when necessary.

The XAI layer provides explainability at two critical levels: **retrieval transparency** and **reasoning traceability**. Retrieval transparency highlights the specific document chunks or sections that contributed to an answer, thereby grounding outputs in verifiable evidence. Reasoning traceability captures the model’s decision pathway, including intermediate logic, feature importance, or interpretability techniques such as SHAP and LIME. Together, these methods enable users to understand not only *what* the system answered, but also *why* and *how* those conclusions were reached.

Complementing explainability, the HITL module introduces a **dynamic oversight mechanism**. Whenever system confidence falls below a defined threshold or guardrails flag inconsistencies, the output is routed to a human reviewer. The reviewer may approve, reject, or correct the AI-generated response, ensuring that critical errors are mitigated in real-time. Furthermore, human feedback is integrated back into the system, strengthening model retraining and continuously improving reliability across use cases.

By combining explainable decision pathways with human validation, the system creates a robust feedback loop that balances automation with expert supervision. This integration not only improves transparency and accountability but also fosters trust in enterprise-level document AI applications where accuracy and auditability are essential.

Keywords — Explainable AI (XAI), Human-in-the-Loop (HITL), Transparency, Trustworthy AI, Document Processing, Multi-Agent Systems.

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LIST OF ABBREVIATIONS

Abbreviation	Description
API	Application Programming Interface
EHR	Electronic Health Record
HITL	Human-in-the-Loop
JSON	JavaScript Object Notation
LLM	Large Language Model
LTM	Long-Term Memory
MIMIC	Medical Information Mart for Intensive Care
MTM	Mid-Term Memory
PubMed	Publicly accessible biomedical literature database, maintained by the US National Library of Medicine
RAG	Retrieval-Augmented Generation
RAGAS	Retrieval-Augmented Generation Assessment System
STM	Short-Term Memory
STORM	Scalable retrieval and generation framework from Stanford, used for robust, context-aware language processing
XAI	Explainable Artificial Intelligence

1. INTRODUCTION

In today’s digital era, enterprises face the challenge of processing massive volumes of unstructured documents such as contracts, invoices, clinical notes, compliance forms, and legal records. Extracting accurate insights from these documents requires Artificial Intelligence (AI) systems that are not only capable of performing information retrieval and extraction, but also ensuring consistency, reliability, and interpretability across complex workflows. Traditional document processing systems often adopt siloed pipelines, where extraction, classification, summarization, and validation are performed independently. This lack of coordination leads to inefficiencies, fragmented outputs, and errors that undermine both user trust and operational scalability [2][8].

Recent advances in **Agentic AI** have introduced a paradigm shift in this field. Instead of relying on a single monolithic model, **multi-agent systems** distribute responsibilities across specialized agents such as retrievers, extractors, validators, and explainers. These agents collaborate under orchestrated workflows, enabling modularity, scalability, and improved adaptability [4][5][8]. Such systems mimic human teamwork—where each agent has a role yet shares information dynamically—resulting in more robust document intelligence. Frameworks such as **Lang Chain** [2] and **CrewAI** [6] provide the infrastructure for orchestrating these workflows, while **context linked memory** ensures knowledge persistence across multiple sessions [1][2].

A central challenge, however, lies in **memory and knowledge sharing**. Without contextual memory, agents often reprocess the same documents repeatedly, leading to inefficiencies and inconsistency. Recent research on **Intrinsic Memory Agents** [1] highlights how structured memory modules—short-term, mid-term, and long-term—can significantly improve agent collaboration and decision-making. Memory-enhanced agents not only retain context but also align outputs across sessions, thereby reducing redundancy and improving reliability.

Another major concern is **hallucination**—factually inaccurate or unsubstantiated AI-generated text. Hallucinations can severely impact decision-making in sensitive domains such as finance, healthcare, or law. Techniques such as **Retrieval-Augmented Generation (RAG)**, schema guided validation, and confidence estimation have emerged as effective strategies to ground outputs in

verifiable evidence and reduce misinformation [5][6][9]. Yet, the integration of these guardrail mechanisms into multi-agent pipelines remains an open research frontier.

Equally important is **explainability and trustworthiness**. Users often require not just correct answers, but also transparency into how these answers were derived. **Explainable AI (XAI)** approaches—including layered prompting and evidence visualization—provide interpretable insights into multi-agent reasoning [7][9]. In high-stakes contexts, **Human-in-the-Loop (HITL)** validation further strengthens oversight, ensuring that AI outputs comply with domain-specific requirements [3][10].

Bringing these strands together, the proposed research envisions a **Modular Multi-Agent Document AI Platform** that integrates:

- **Contextual Memory & Data Engineering** – Hierarchical memory combining vector databases with semantic retrieval to support persistence and context sharing [1][2][6].
- **Multi-Agent Orchestration** – A role-based framework coordinating extractors, retrievers, validators, and explainers in collaborative workflows [4][5][8].
- **Guardrails & Hallucination Mitigation** – Evidence-grounding, schema validation, and confidence estimation mechanisms to minimize misinformation [5][6][9].
- **Explainable AI & HITL** – Transparency layers for auditability and human oversight, enhancing trust in AI-driven document processing [3][7][9][10].

By combining these components, the system will deliver reliable, auditable, and context-aware document intelligence. It addresses the critical research gaps in memory integration, hallucination control, and explainability, while advancing the state of multi-agent collaboration in real-world enterprise workflows [1][2][4][5][6][7][8][9][10].

1.1 Background and Literature Survey

In multi-agent document AI systems, explainability and human oversight are essential to ensuring transparency, trust, and accountability. While contextual memory and retrieval mechanisms enhance continuity and factual grounding, they do not inherently guarantee that outputs are interpretable or aligned with organizational and regulatory requirements. Without visibility into how outputs are derived—or the ability for humans to intervene when confidence is low—systems risk producing opaque, error-prone, or even unsafe results [1][5][9].

Explainable AI (XAI) provides mechanisms for making multi-agent reasoning processes transparent at both the retrieval and generation levels. For example, retrieval transparency highlights which document chunks or sections contributed to an answer, enabling users to directly verify the evidence supporting a response. Reasoning transparency further traces the decision-making path of agents, revealing how an extracted entity, classification, or summarization was derived. Techniques such as SHAP and LIME have shown promise in providing feature-level importance scores for classification tasks, while prompt-based reasoning traces allow large language models (LLMs) to expose intermediate decision steps [1][8]. Together, these methods support auditability and help establish trust in enterprise workflows.

Human-in-the-Loop (HITL) systems complement XAI by introducing an active oversight mechanism. Rather than treating AI decisions as final, HITL architectures route uncertain or noncompliant outputs to human reviewers for verification, correction, or enrichment [7]. This ensures that mission-critical processes—such as compliance checks, financial auditing, or medical record analysis—maintain a high level of reliability. Moreover, HITL not only mitigates risk but also closes the feedback loop: user corrections can be logged, structured, and reintegrated into contextual memory or retraining datasets, thereby continuously improving agent performance over time [6][7].

Recent studies emphasize the synergy between explainability and human oversight. For instance, Wang et al. [6] highlight how transparent reasoning traces combined with reviewer feedback substantially reduce model drift in long-running enterprise workflows. Similarly, Amershi et al. [7] show that HITL systems significantly improve both user trust and model adaptability, particularly

in high-stakes environments where full automation remains risky. By integrating these principles, multi-agent systems move beyond “black-box” automation and toward accountable, human-aligned intelligence.

Taken together, XAI and HITL form a dual layer of reliability within modular multi-agent frameworks. XAI ensures interpretability and evidence-backed transparency, while HITL provides human safeguards against hallucination, low-confidence outputs, and domain-specific edge cases. This combined layer enables scalable yet trustworthy deployment of document AI across regulated industries such as law, healthcare, and finance [1][6][7][8][9].

1.2 Research Gap

Despite progress in explainable AI (XAI) and Human-in-the-Loop (HITL) systems, several critical gaps remain when applied to agentic AI for enterprise document automation:

- **Limited Multi-Agent Explainability** – Most current XAI research focuses on explaining single-model predictions. In document automation pipelines, however, multiple specialized agents (extractor, validator, verifier) collaborate to produce results. Existing methods rarely provide *end-to-end traceability across agents*, making it difficult to understand how different decisions interact to shape the final output.
- **Shallow Explanations for Complex Decisions** – Current XAI approaches often highlight evidence or attention weights, but they do not adequately capture *why an agent discarded alternatives* or *how conflicting evidence was resolved*. This reduces transparency in high stakes document workflows where multiple interpretations are possible.
- **Weak Integration of XAI with HITL** – While HITL frameworks exist, they are typically reactive (humans only intervene after low-confidence predictions). Few systems provide *proactive, explanation driven HITL*, where humans are guided by interpretable rationales, alternative candidates, and agent reasoning traces to make informed interventions.
- **Lack of Iterative Feedback Loops** – Explanations are often static and do not evolve after human feedback. Current pipelines rarely integrate HITL feedback into the XAI layer to *improve future explanations* or *adapt agent behaviors dynamically*. This leads to repetitive errors and undermines long-term trust.

- **Absence of Standardized Evaluation** – There is no established framework for measuring the effectiveness of explanations in multi-agent, HITL-enabled document systems. Most evaluation metrics (e.g., fidelity, interpretability) are defined for single models, leaving a gap in assessing how well explanations support *collaborative decision-making between AI agents and humans*.

These gaps highlight the need for **a unified framework that couples multi-agent reasoning with interactive XAI and HITL feedback loops**, enabling traceable, transparent, and adaptive document automation in enterprise settings.

1.3 Research Problem

Enterprise document workflows require AI systems that can perform complex reasoning, maintain accuracy, and adapt across multiple interactions, while ensuring transparency, compliance, and trustworthiness. Current solutions often fall short—they provide outputs without clear explanations, fail to involve human oversight effectively, and treat extraction, reasoning, and validation as separate processes, which limits their reliability and accountability [2][3][5].

This research seeks to answer the following question:

How can we design an agent-aligned, explainable, and human-in-the-loop document AI platform that provides interpretable, evidence-backed outputs, integrates human feedback effectively at critical decision points, and optimizes system performance for efficiency, latency, and accuracy, enabling safe, auditable, and trustworthy document understanding?

By addressing this problem, the proposed system will allow each agent in the pipeline to operate with role-specific responsibilities while leveraging XAI techniques to provide interpretable outputs and HITL mechanisms to incorporate human validation and correction [2][3][5]. This approach promises more reliable, coherent, and accountable document processing, ultimately supporting scalable, transparent, and context-aware AI systems for enterprise environments [6].

2. OBJECTIVES

2.1 Main Objective

The main goal of this research is to design and implement a comprehensive multi-agent document AI system capable of efficiently processing large volumes of unstructured documents. Unlike traditional systems that handle tasks in isolation, the proposed platform will maintain context across multiple interactions, allowing agents to collaborate effectively while retaining relevant knowledge. The system integrates specialized agents for tasks such as extraction, retrieval, verification, and explanation, coordinated through a modular and scalable framework. At the heart of the platform is a hierarchical memory module that enables agents to store and retrieve information across short-term, mid-term, and long-term workflows, ensuring continuity and reducing redundant processing. Retrieval mechanisms will be closely integrated with evidence grounding to enhance factual accuracy, while verification layers and guardrail mechanisms ensure outputs are reliable and compliant with organizational or regulatory standards. An explainability component will provide transparency into the system's reasoning, allowing users to understand and trust AI-generated insights.

Overall, the proposed system aims to provide a flexible, scalable, and auditable document AI platform that can adapt to complex enterprise workflows, improve efficiency, and support reliable decision-making across diverse domains such as law, finance, and healthcare.

2.2 Sub Objectives

Design and Development of Explainable AI Components

The first sub-objective focuses on embedding explainability into every stage of the multi-agent workflow. This involves designing interpretable reasoning paths where each agent—extraction, verification, or summarization—provides evidence-backed outputs with transparent justifications. The system will generate human-readable explanations that clarify *why* a decision or extraction was made, thereby improving trust, accountability, and compliance in enterprise document processing.

Implementation of Human-in-the-Loop (HITL) Mechanisms

The second sub-objective emphasizes structured human oversight at critical decision points. Domain experts will be able to review, validate, and correct AI outputs, especially in high-stakes contexts such as legal contracts or financial reports. These interventions will be systematically integrated into the workflow, ensuring human judgment complements automation while reducing risks of errors and bias.

Integration of Grounding and Provenance Tracking

A critical sub-objective is the tight integration of retrieval-augmented generation with provenance tracking and explainability. Every retrieval and AI-generated output will be supported with verifiable sources, schema validation, and domain-specific rules. By coupling grounding with XAI, the system ensures that not only are outputs factually consistent, but users also understand *where* and *how* the information was derived, strengthening auditability in regulated domains.

Continuous Learning from Human Feedback

The fourth sub-objective focuses on embedding feedback loops where expert corrections are not only applied immediately but also used to refine future system behavior. This creates a self-reinforcing cycle where HITL validation improves short-term accuracy and long-term model performance. Over time, the system will adapt to organizational norms and domain-specific knowledge, becoming more aligned with expert decision-making processes.

Adaptive Explainability and Oversight Policies

Finally, the system will implement adaptive strategies for balancing explanation detail, human oversight, and performance. For routine or low-risk tasks, concise explanations may suffice, while for high-stakes or regulated workflows, more detailed reasoning traces and mandatory HITL approval will be required. This adaptive policy design ensures efficiency without sacrificing compliance, scalability, or trustworthiness.

3. METHODOLOGY

3.1 System Architecture

User Input Layer

Users provide documents (PDF, Word, text, JSON) or queries through a user interface. The input is passed via an API to an **Orchestrator Agent**, which determines the processing path: structuring the document or generating an answer. This layer provides the entry point for both AI processing and human oversight.

HITL Workflow

The **HITL interface** ensures that human experts can intervene in decision-making where AI confidence is low, or outputs are flagged as non-compliant:

- **Review and Validation:** Human reviewers examine AI outputs flagged for uncertainty, inconsistency, or regulatory concern.
- **Correction and Feedback:** Corrections are applied to structured data or answers. These updates are fed back into the system to refine future predictions.
- **Continuous Learning Loop:** Feedback updates both the procedural logic of agents and the vector database, reducing repeated errors and improving contextual understanding over time.

HITL Workflow Steps:

1. Document processed by agents (extraction, classification, summarization).
2. Output passes through **guardrail and hallucination checks**.
3. Low-confidence or non-compliant outputs are routed to human experts.
4. Human corrections are stored and used to update the system knowledge base, improving subsequent AI predictions.

Explainable AI Layer

The XAI component provides transparency and traceability for all AI-generated outputs:

- **Retrieval Transparency:** Shows which document sections, paragraphs, or facts were used to generate the output (e.g., “This answer is based on Page 3, Paragraph 2”).
- **Reasoning Trace:** The AI generates interpretable explanations of its reasoning, highlighting the contribution of each input feature to decisions. Tools such as **SHAP** or **LIME** can be applied to provide additional interpretability for structured outputs and classification tasks.
- **Integration with HITL:** Explanations support human reviewers in understanding AI decisions and provide context for more accurate corrections and feedback.

System Integration and Feedback Loop

The combination of HITL and XAI enables a **closed-loop system** for continuous improvement:

1. AI generates predictions with associated reasoning and confidence scores.
2. XAI layer presents explanations to human reviewers.
3. HITL feedback is incorporated into the memory and vector database.
4. Agents adapt over time, improving reliability, interpretability, and compliance.

By integrating HITL and XAI, the methodology ensures that enterprise document workflows remain **auditable, transparent, and continuously improving**, while reducing errors and increasing trust in AI outputs.

By combining structured, agent-specific memories with factual grounding and human oversight, this methodology creates a robust contextual memory system that enhances cooperation between agents, reduces errors, and supports safe and auditable document AI at scale.

3.2 System Diagram

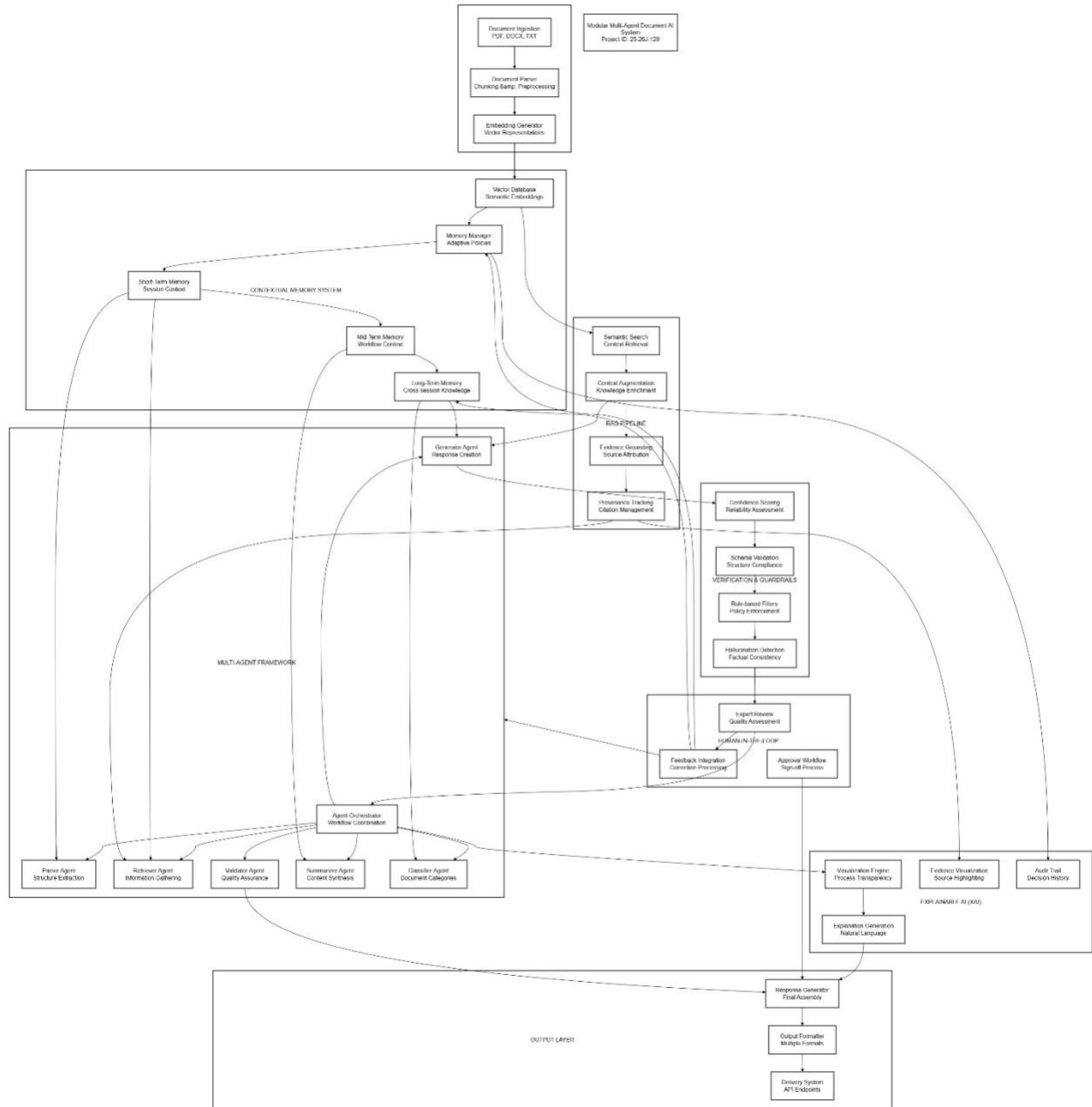


Figure 1

4. PROJECT REQUIREMENTS

4.1 Functional Requirements

Explainable AI (XAI)

- The system shall provide **retrieval transparency** by displaying which document segments or chunks were used in generating an output.
- The system shall generate a **reasoning trace** that explains how retrieved evidence supports the final answer or decision.
- The system shall highlight influential features or text passages that contributed to a classification, extraction, or summarization outcome.
- The system shall provide **multi-level explainability**, with concise summaries for end-users and detailed reasoning paths for expert reviewers.
- The system shall integrate XAI outputs with the user interface, enabling reviewers to inspect sources, logic, and decision pathways.

Human-in-the-Loop (HITL)

- The system shall route outputs to human reviewers when confidence levels fall below a defined threshold or when outputs fail guardrail checks.
- The system shall allow human experts to **approve, reject, or correct** AI-generated outputs.
- The system shall store human feedback in both structured knowledge bases and vector memory for continuous system improvement.
- The system shall ensure **auditable tracking** of all human interventions, including reviewer identity, timestamp, and applied corrections.
- The system shall enable **feedback loops**, where human corrections are used to refine embeddings, retraining data, and agent logic.
- The system shall support **adaptive oversight**, requiring more detailed HITL review for high-risk or regulated documents while minimizing unnecessary intervention in low-risk cases.

4.2 Non-functional Requirements

Performance

- Retrieval latency shall be minimized for document sets.
- The system shall handle queries across tens of thousands of document chunks efficiently.

Scalability

- The system shall scale horizontally by supporting distributed vector database storage.
- The design shall allow integration of additional memory layers without major redesign.

Reliability

- The system shall ensure consistent retrieval results for identical queries when context remains unchanged.
- The system shall handle partial failures gracefully (e.g., fallback retrieval if one database node fails).

Security

- All stored embeddings and metadata shall be encrypted.
- The system shall enforce access control for sensitive enterprise documents.

Maintainability

- The system shall provide modular APIs for ingestion, retrieval, and memory management.
- The system shall allow easy replacement of embedding models without redesigning the pipeline.

Extensibility

- The system shall support plugging in new vector database backends or embedding models with minimal changes.
- The system shall allow future extension to integrate feedback-based learning.

Usability

- The memory retrieval outputs shall be formatted in a standardized JSON schema for downstream agents.
- The system shall expose clear API endpoints for developers to integrate with the orchestrator.

Auditability

- The system shall log retrieval operations with timestamps, source documents, and relevance scores.
- The system shall support provenance tracking of retrieved context for compliance and audit needs.

4.3 Expected Test Cases

Test Case ID	Test Case Description	Pre-Conditions	Test Steps	Expected Result
TC-01	Retrieval transparency	System running, documents ingested & embedded	Submit a query, Run retrieval + answer generation, Request explanation	System displays the exact document chunks (page/paragraph IDs) used in generating the answer
TC-02	Reasoning trace generation	System running with reasoning trace enabled	Submit query requiring classification/decision, Generate reasoning trace	System outputs reasoning steps showing how evidence led to the decision (e.g., extracted date, compared with today, expired)
TC-03	HITL escalation on low confidence	Confidence threshold configured (e.g., 0.75)	Submit ambiguous query, run pipeline, Check confidence score	If confidence < threshold, output is routed to human reviewer interface
TC-04	HITL feedback loop integration	Reviewer access available, vector populated DB	Submit query, Generate AI output, Reviewer corrects data. Submit correction	Embeddings stored with Correction stored in vector DB/knowledge base and reflected in future outputs

Table 1

5. BUDGET AND BUDGET JUSTIFICATION

The table below depicts the budget required for the entire proposed system with other related payments.

Description	Cost in LKR
Cloud deployment + Computing power	100000
Conference participation and related travel.	40000
Software licenses	25000

Table 2

6. COMMERCIALIZATION

Target Industries

The system targets healthcare, legal, finance, and education, where secure and intelligent document processing is critical. Features like contextual memory, HITL feedback, hallucination detection, and guardrails ensure reliability and trust.

Revenue Model

A subscription-based, tiered model will serve SMEs, corporates, and research institutions. Enterprise clients get customized deployments and premium support. API licensing provides an additional revenue stream.

Go-to-Market

Pilot projects with leading organizations, case studies, digital campaigns, conferences, and cloud-provider partnerships will drive credibility and market reach.

Intellectual Property

Protection will be ensured through software copyrights and patents on multi-agent contextual memory.

Continuous Improvement

User feedback, domain-specific modules, and integration with new AI frameworks will keep the system evolving.

Long-Term Expansion

Expansion begins with Asian markets, then global networks, supported by collaborations with universities and research institutes.

7. CONCLUSION

The proposed modular multi-agent document AI system presents a new and impactful solution to the growing problems of unstructured document processing. By combining contextual memory, HITL feedback, hallucination detection, and guardrails, the system enables accuracy, dependability, and scalability across industries such as healthcare, finance, law, and education. Aside from its technical strengths, the project highly values real-world applicability through a sustainable commercialization strategy, protection of intellectual properties, and long-term growth plans.

Ultimately, this solution not only addresses pain points in the industry today but also positions a base for future innovation and adaptability in a changing AI-driven world. With its capacity for optimizing efficiency, compliance, and decision-making, the proposed system is well-positioned to create significant commercial and societal value.

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